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Experiment 4

Aim: Write a program to simulate Election algorithm (Bully algorithm).

Theory:

- In a distributed systems, multiple processes run independently. When the coordinator process fails, an election algorithm is used to select a new leader.

• **Bully Algorithm**

- The Bully algorithm selects the process with the highest ID as the leader.
- If the current leader fails, the next highest process takes over.
- The algorithm assumes each process knows other processes IDs.
- the process with the highest ID always wins the election.
- It is called a bully algorithm because a higher-~~ID~~ process can "bully" lower-ID processes by forcing them to accept its leadership.

1) **Detecting Failure**

⇒ A process detects that the leader has failed. The process initiates an election if its ID is higher than the failed leader.

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2) Election process

⇒ The initiating process sends an "ELECTION" message to all higher-ID processes.

- If a higherID process is alive, it respond with an "OK" message and starts its own election.
- If no higher-ID process replies, the initiating process declares itself the leader.

3) Declaring the New leader

⇒ The new leader broadcasts a "COORDINATOR" message to inform all processes.

- All other process update their records to recognize the new leader.

- For e.g.,

Given 5 processes, P_1, P_2, P_3, P_4, P_5
(Assume P_5 is current leader and fails)

Process	Action
P_2 detects failure	starts an election.
$P_2 \rightarrow P_3, P_4$	Sends "Election" message
P_3, P_4 reply with "OK"	They start their own election
$P_4 \rightarrow P_5$ (No response)	P_4 declares itself leader.
P_4 broadcasts "COORDINATOR"	All processes recognize P_4 as the new leader.

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